ALEXANDER IMMANUEL GOERKE ZAMORA

Valencia, Spain | alexander.goerke2004@gmail.com | +34 635 426 375
aigoezam.github.io | linkedin.com/in/alexanderimmanuelgoerke

SUMMARY

Industrial Electronics and Automation engineer from UPV, with an Erasmus+ year at DTU (Copenhagen). My focus is robotics, digital control, and signal processing, from theoretical derivation through simulation to real hardware implementation and experimental validation, in international and multidisciplinary environments.

EDUCATION

IALE INTERNATIONAL SCHOOL — **Valencia, Spain**
Secondary Education (ESO & Bachillerato) — *2016 – 2022*
- Awarded the Science Prize for academic excellence.

UNIVERSITAT POLITÈCNICA DE VALÈNCIA (UPV) — **Valencia, Spain**
B.Sc. Industrial Electronics and Automation Engineering — *2022 – 2026*

TECHNICAL UNIVERSITY OF DENMARK (DTU) — **Lyngby, Denmark**
Erasmus+ Exchange — *2025 – 2026*

EXPERIENCE

CEBREO MEDICAL A/S — **Copenhagen, Denmark**
Software Engineer — *Jan 2026 – Jun 2026*
- Built EEG signal viewers with filtering, accelerometer motion visualisations, and PPG pulse-analysis tools to support clinical biosignal inspection across EDF, CSV, and .mat data formats.
- Developed synthetic test-signal pipelines for algorithm validation and QA, within a multidisciplinary team of software engineers, physiologists, and product developers.

PRIVATE TUTORING — **Valencia, Spain**
Maths, Physics & Chemistry Tutor — *2022 – Present*
- One-to-one instruction for secondary school students; designed study plans and applied structured problem-solving to improve academic outcomes.
- Recommended by school teachers following strong academic results; developed clear communication and mentoring skills.

PROJECTS

BACHELOR THESIS – DELTA ROBOT — **DTU / UPV, 2025–2026**
- Designed the control system for a delta robot for a 5-year PhD programme at DTU on agricultural weed-intervention; derived full IK/FK kinematics and built a complete simulation suite in Python, then ported the controller to C++ and deployed it on Teensy 4.1.
- Validated feedforward, smooth bounded feedback, and full PID strategies, achieving 1–2 mm end-effector positioning accuracy through quantitative real-robot experiments.

DIFFERENTIAL-DRIVE ROBOT — **DTU Digital Control, 2025**
- Ranked 1st in class; designed PI/PD controllers with encoder-based sensor fusion, validated in Simulink and deployed on Teensy 4.1.

VISION-GUIDED 4-DOF ROBOT — **DTU Robotics, 2025**
- Built a robot to detect objects, convert pixel coordinates to world space, and reach them via inverse kinematics; implemented HSV segmentation, camera calibration, and Dynamixel servo control.

TECHNICAL SKILLS

- **Engineering:** Robotics, Digital Control, Signal Processing, Embedded Systems, Circuit Design, IK/FK Kinematics, Sensor Fusion, Mechatronics
- **Programming:** Python, C/C++, MATLAB, Arduino, HTML/CSS
- **Design & Simulation:** SolidWorks, AutoCAD, MATLAB/Simulink, Proteus, OrCAD
- **Tools & AI:** Git, VSCode, LaTeX, Claude, Codex

CERTIFICATIONS & AWARDS

- ISO 21500 & ISO 21502: Project, Programme and Portfolio Management (Danish Standards, Jan 2026)
- Advanced English (C1) | German (A2) | Web Programming (HTML/CSS/JS) | Energy Systems (CFP–UPV, 2024)

Languages: Spanish (native) | English (C1) | German (A2)